DO NOW — RECALL

# Fetch–decode–execute and buses

Name: _____ Class: _____ Date: _____

**1. Which register holds the address accessed during a fetch?**

**2. What does the CIR hold?**

**3. What kind of information travels along the address bus?**

**4. A rising-edge D flip-flop has Q=0. D changes to 1 between clock edges; the next rising edge occurs while D=1. State Q before and after that edge, and explain why.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

MAR.

### 2. Model answer

The current instruction that is being decoded or executed.

### 3. Model answer

Addresses identifying the memory or device location.

### 4. Model answer

Before edge Q stays 0; at rising edge D=1 is sampled so Q becomes 1. It changes on the active edge, not merely when D changes.

### Vocabulary — know these terms

fetch — Retrieve next instruction from memory.

decode — Interpret opcode and operands.

execute — Perform the operation.

address bus — Transfers memory/device addresses.

data bus — Transfers data and instructions.

control bus — Transfers signals coordinating reads and writes.

NEW LEARNING

# Read, annotate and explain

### Fetch uses the buses

The address bus carries the address selected by MAR; memory returns instruction data on the data bus; control signals coordinate the transfer.

### Decode interprets instruction

The control unit examines CIR to decide which operation and operands are involved.

### Execute changes state

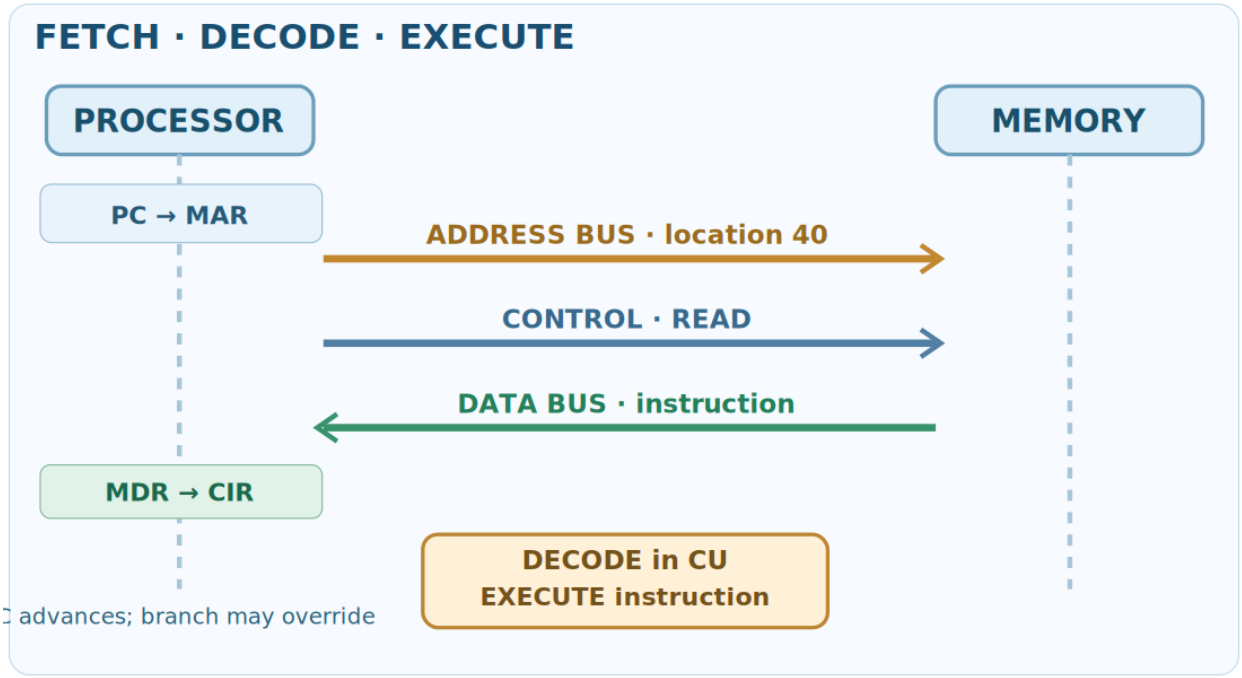
An instruction may update ACC or memory, or alter PC for a jump. Interrupt handling is coordinated at appropriate points and can temporarily redirect execution.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

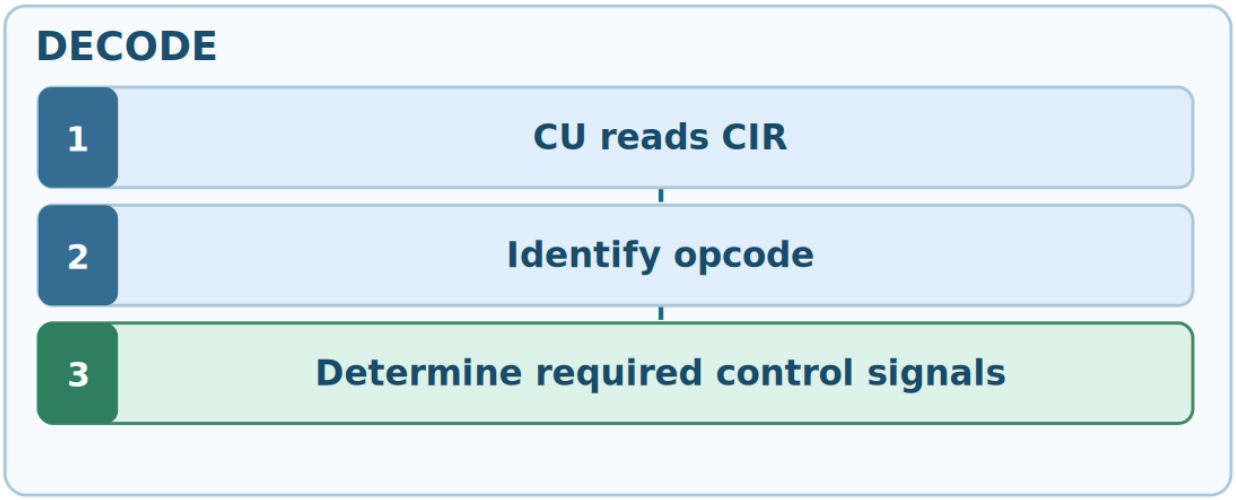
Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Fetch-decode-execute sequence showing CPU and memory with address bus CPU to memory, read control signal, data bus memory to CPU and instruction register and PC updates.

### New Learning

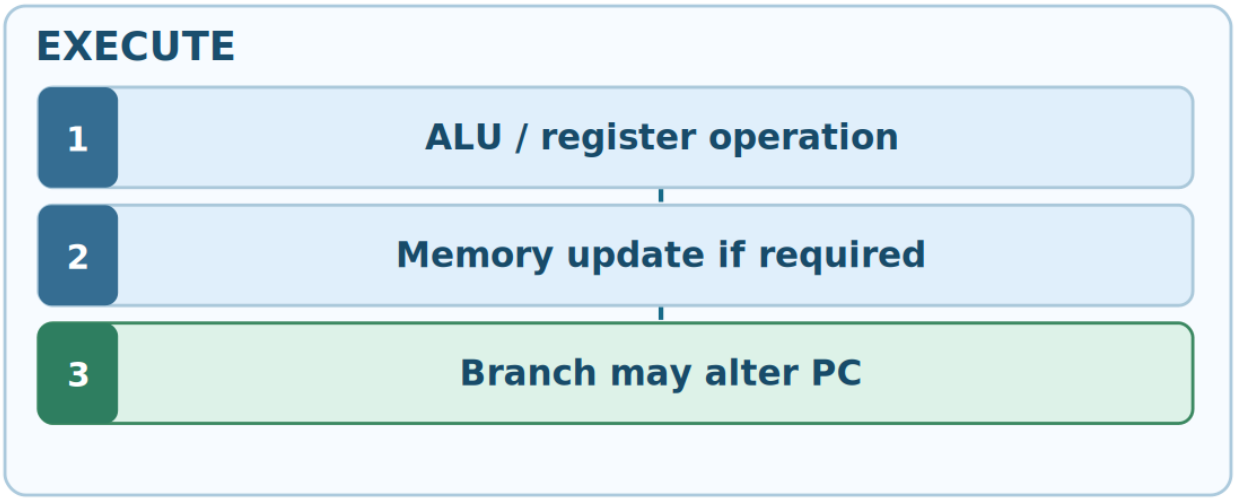



DECODE; CU reads CIR; Identify opcode; Determine required control signals

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




EXECUTE; ALU / register operation; Memory update if required; Branch may alter PC

I DO — WORKED EXAMPLE

# Follow the worked method

### Fetch the instruction at address 40 and explain what each bus contributes.

### Predict first

MAR identifies location 40; the address bus carries that location to memory, accompanied by a read control signal.

### Explain the answer

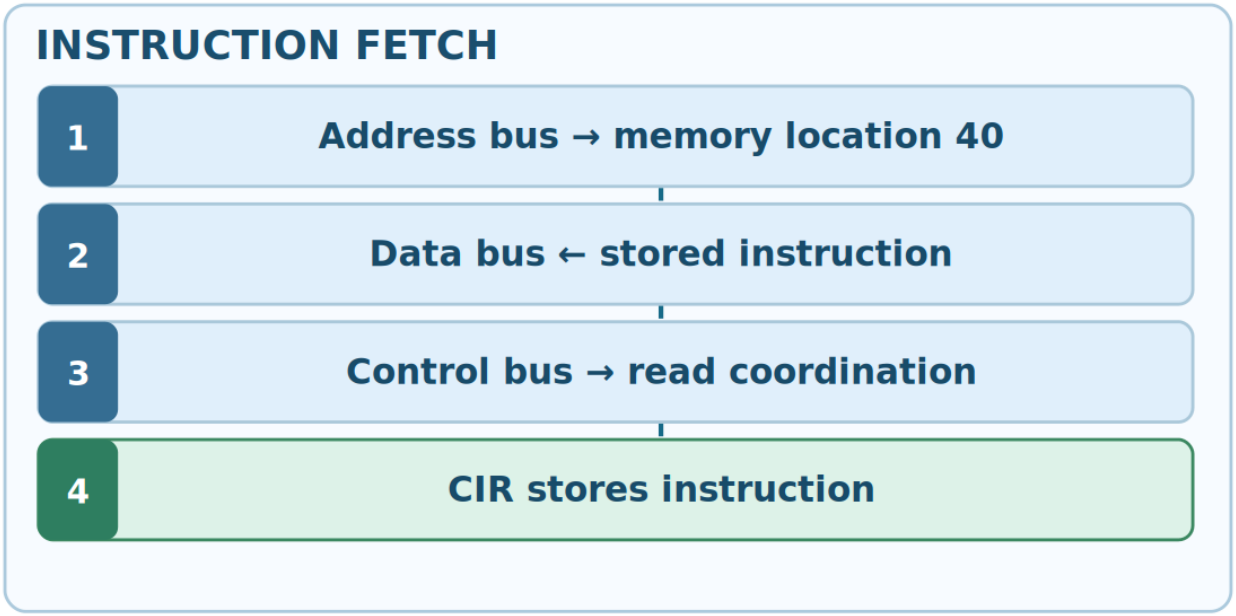
The data bus transfers instruction content to MDR, then CIR; sequential PC becomes 41.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Fetch the instruction at address 40 and explain what each bus contributes.**

MAR identifies location 40; the address bus carries that location to memory, accompanied by a read control signal.




INSTRUCTION FETCH; Address bus → memory location 40; Data bus ← stored instruction; Control bus → read coordination; CIR stores instruction

WE DO — GUIDED PRACTICE

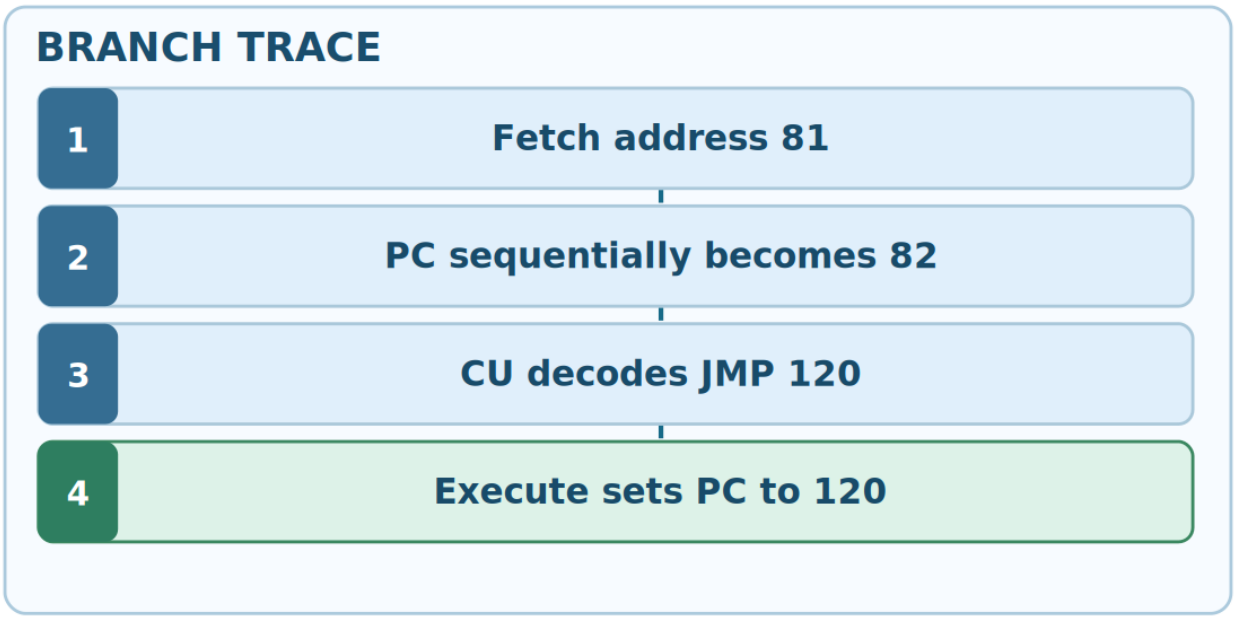
# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**An instruction at address 81 is an unconditional jump to 120. What happens to PC?**

The initial fetch advances PC from 81 to 82, as in a sequential instruction.




BRANCH TRACE; Fetch address 81; PC sequentially becomes 82; CU decodes JMP 120; Execute sets PC to 120

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

**Guided 1: Predict first The initial fetch advances PC from 81 to 82, as in a sequential instruction.**

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

For each bus—address, data and control—state its main function and give one example of the information transported.

### Question 2

Trace a fetch starting PC=64, then explain what happens when CIR contains JMP 100.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Compare what happens in the instruction cycle during a normal arithmetic instruction and while responding to an interrupt.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**An unconditional branch can change the Program Counter away from its sequential value.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Explain the answer

Executing the unconditional branch overwrites PC with target address 120. The next fetch occurs there, not at 82.

IMPROVE: Explain one correction to your own reasoning.